PREDICTING YOUTH SUBSTANCE USE USING DECISION TREES

DATA 5322- Statistical Machine Learning -2

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INTRODUCTION

Data: NSDUH 2023

- National Survey on Drug Use and Health
- 10,561 respondents under age 18

The three questions being answered are:

- Did this youth ever use alcohol? (Binary)
- How frequently do marijuana users consume it? (Multi-class)
- How many days per year is marijuana used? (Regression)

THEORETICAL BACKGROUND (DECISION TREES)

- Recursively splits data by the feature that best reduces impurity
- Classification uses Gini index, Regression uses RSS
- Results in an interpretable flowchart of decisions

Tuning: Tree depth is controlled using cross-validation pruning, cost-complexity parameter aplha

Evaluation: Accuracy / confusion matrix for classification, RMSE for regression

Limitation: Single trees are high variance and prone to overfitting

THEORETICAL BACKGROUND (ENSEMBLES)

Model	How it works	Key Parameter	Limitation
Bagging	Bootstrap + average many trees	mtry = all features (p)	Trees are correlated
Random Forest	Bagging + random feature subset per split	mtry = √p	Less interpretable
Gradient Boosting	Sequentially corrects errors of prior trees	n.trees, shrinkage, depth	Sensitive to tuning

METHODOLOGY: DATA PREPARATION

Variable Selection:

- 16 predictors chosen from peer, parental, school, and demographic categories
- Avoided leakage an example is did not use "age first used alcohol" to predict "ever used alcohol"

Cleaning Steps:

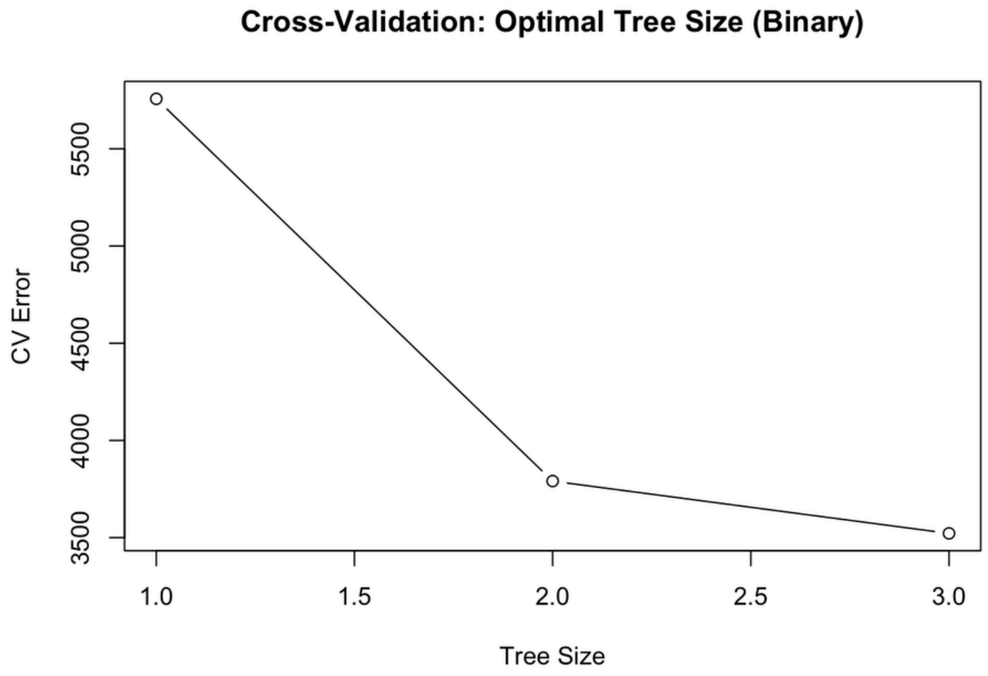
- Removed special coded non-answers (991 = never used, 993 = not used recently)
- Converted all categorical variables to factors
- Multi-class filtered to marijuana users only (MRJFLAG = 1)

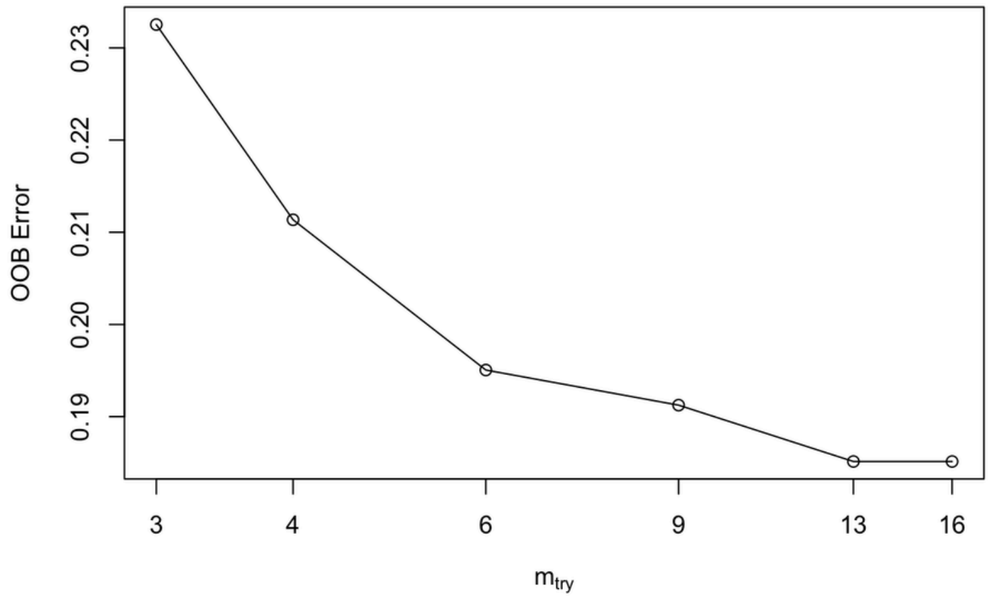
Class Imbalance:

- Binary outcome: (77% No, 23% Yes) addressed using ROSE oversampling
- Balanced training set: 50/50 split

Train/Test Split: 80% training, 20% testing across all three

HYPERPARAMETER TUNING





HYPERPARAMETER TUNING

Decision Tree Pruning:

- Used cross-validation misclassification error
- Binary: optimal size = 3 nodes
- Regression: optimal size = 3 nodes

Random Forest mtry:

- Used tuneRF with stepFactor = 1.5
- Searched mtry values: 3, 4, 6, 9, 13, 16
- OOB error decreased with higher mtry

Gradient Boosting:

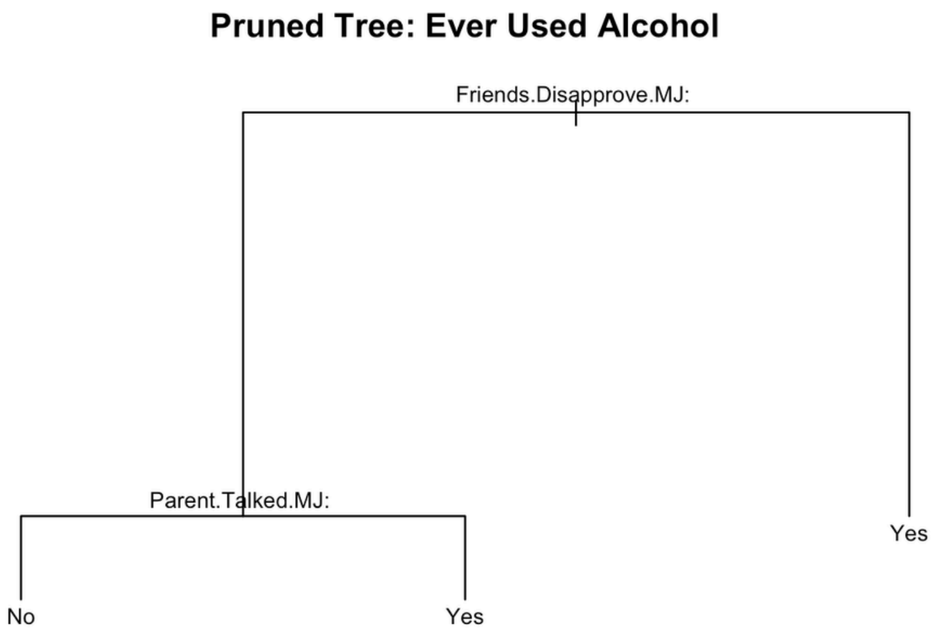
- GBM: Joint grid search over depth (1–4) and shrinkage (0.001–0.1), best trees via 5-fold CV

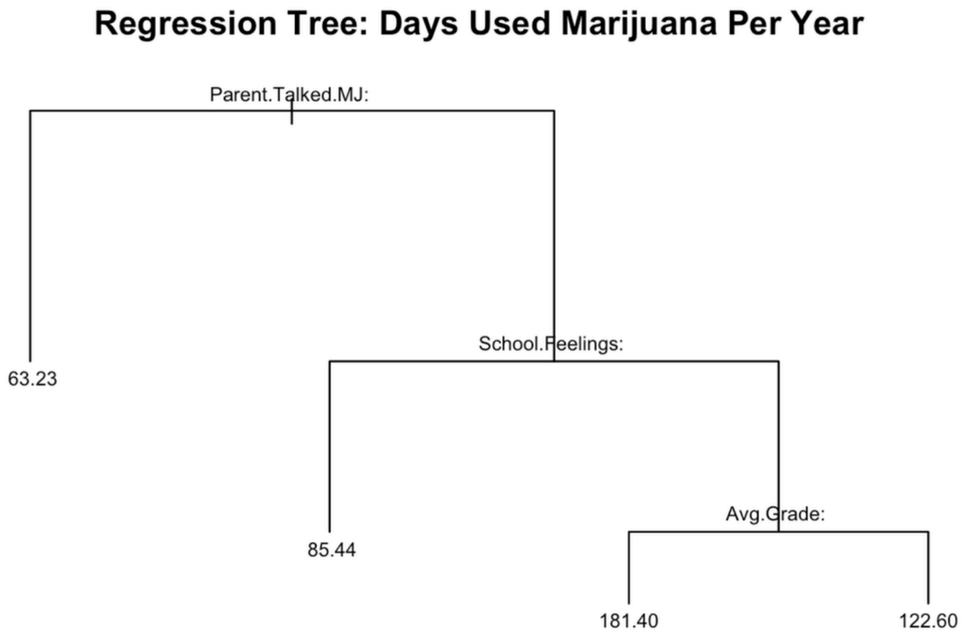
ENCODING COMPARISON

Encoding Type	Variable	What it measures	Best for
Binary (0/1)	ALCFLAG	Ever used alcohol or not	Simple yes/no classification
Ordinal factor	MRJYDAYS (1–6)	Ordered frequency categories	Multi-class when order matters
Continuous integer	IRMJFY (1–365)	Exact days of marijuana use per year	Regression when precision needed

Encoding	Accuracy
Nominal	74.10%
Ordinal	73.70%
Numeric	74.80%

TREE INTERPRETATION





FRDMEVR2 = Friends disapprove of marijuana use
PRMJEVR2 = Parent told youth marijuana is wrong
SCHFELT = How youth feels about school
AVGGRADE = Average grade in school

RESULTS: BINARY CLASS

Model	Accuracy
Basic simple Tree (imbalanced)	77.20%
Balanced Tree	73.60%
Pruned Tree	73.60%
Bagging	70.80%
Random Forest	**73.90%**
Boosting	71.90%

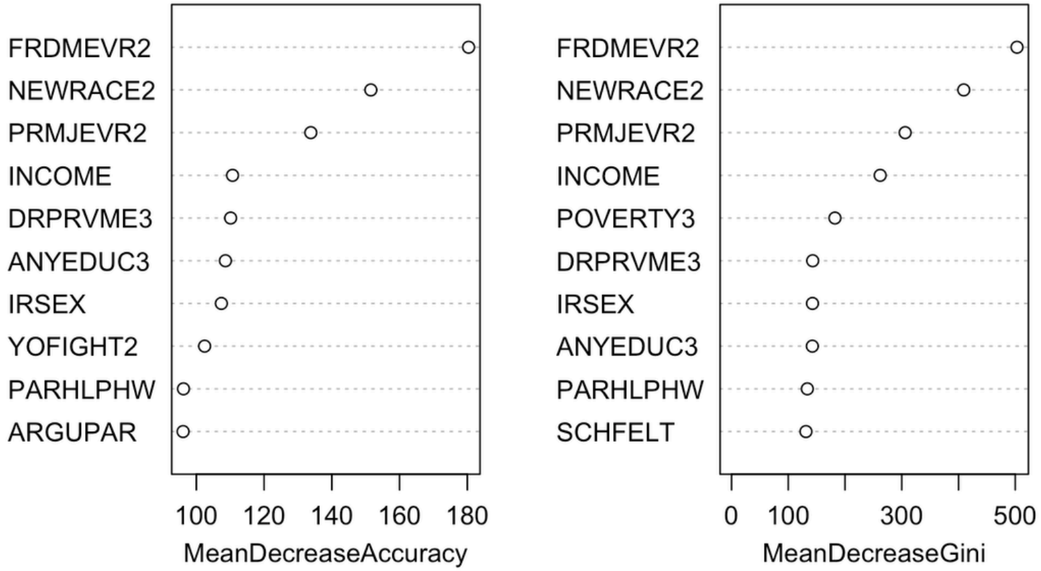
RESULTS: MULTI-CLASS AND REGRESSION

Multi-Class	
Model	**Accuracy**
Basic Tree	23.30%
Pruned Tree	23.30%
Bagging	25.30%
Random Forest	24.70%
Boosting	**29.10%**

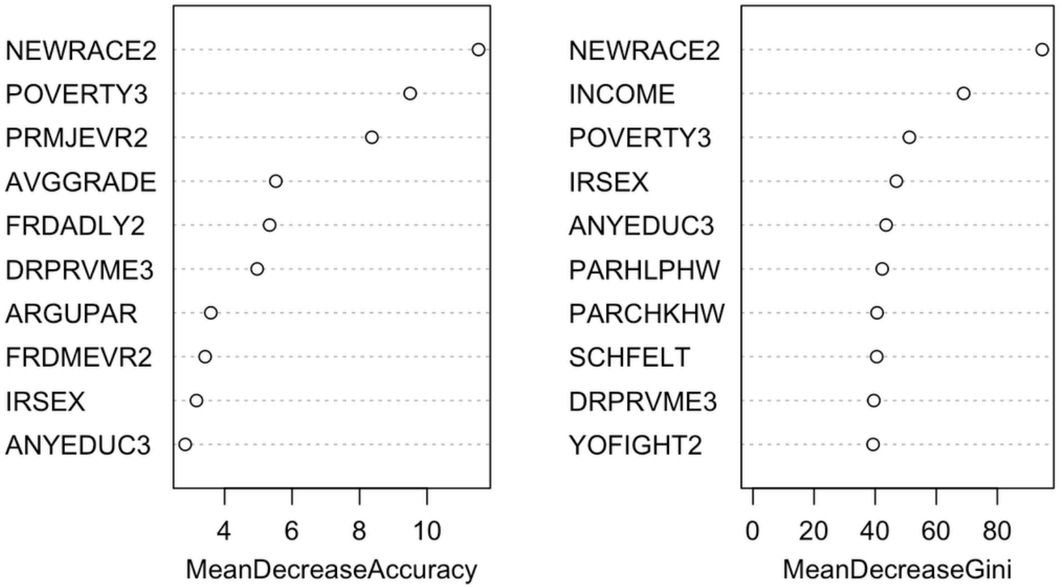
Regression	
Model	**RMSE**
Basic Tree	**109.6**
Pruned Tree	112.6
Boosting	110.3
Random Forest	111.4
Bagging	117.5

VARIABLE IMPORTANCE

Variable Importance of Alcohol



Variable Importance of Marijuana Frequency



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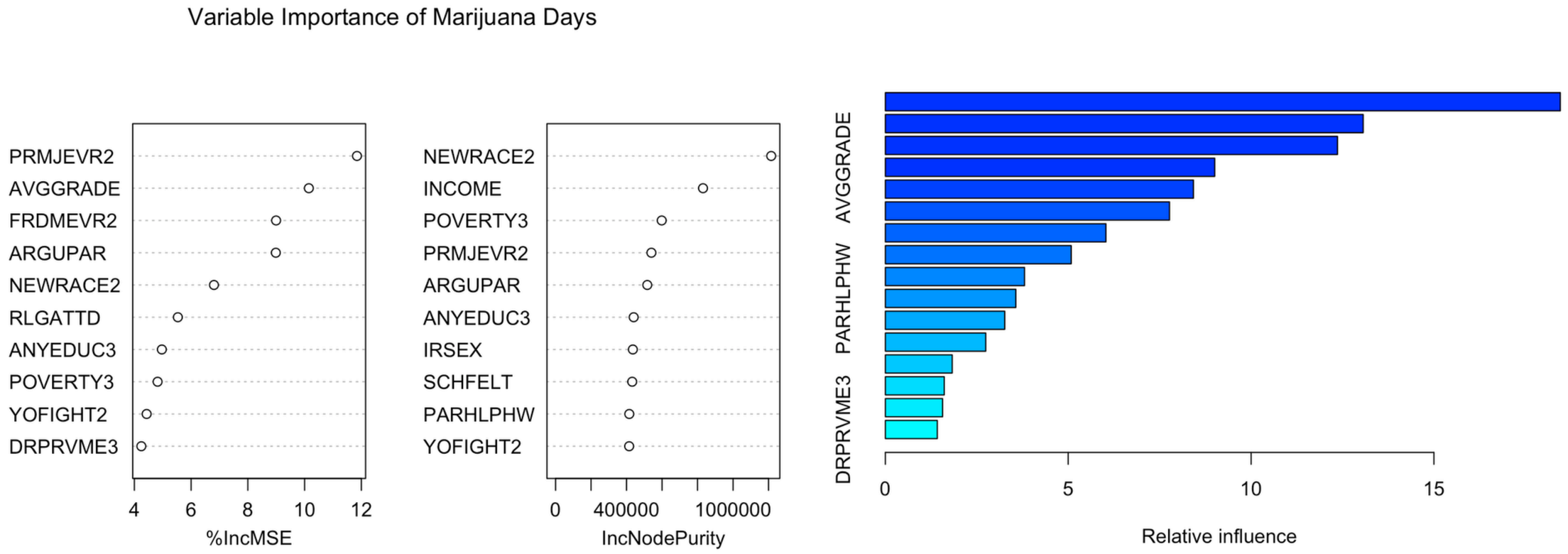

DISCUSSION

Performance of Model:
Random Forest most effective for binary (73.9%), Boosting most effective for multi-class (29.1%)
Basic tree proved quite competitive for the regression problem but not always superior

Reasons behind poor performance of model:
Multiclass classification: 6 overlapping frequency bins difficult to classify based on survey responses
Regression problem: Survey respondents can report days between 1-365

Limitations:
Response bias in the survey data
Binary nature of most features, low granularity
Correlation but no causation
Race and income as features indicative of systemic issues

Future Work:
Include psychological feature
SMOTE technique for balance over ROSE

CONCLUSION

- Family and peer influences have greater predictive power than mere demographics
- Simple trees had grasped the important patterns, ensemble methods helped with stability, but not much more
- The prediction of frequency, the multi-class is much more difficult than a binary classification
- Drug use behavior is very complicated, and this is just one step toward understanding the problem
- Implication: Preventive measures that emphasize family and peer environments may prove more successful than mere demographics

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THANK YOU